CSP RECENT EXAM RECALL - MASTER REVISION GUIDE

Deduplicated recalled topics | CSP-style practice questions | Correct option shown entirely in red | Short complete recall notes

How to use: Attempt each question before looking at the red option. Then read the recall note. Where the original recollection was incomplete or ambiguous, the reconstructed question is clearly written to test the supported concept rather than pretending the original wording is known.

Source basis: Your recent-exam recall files and CSP study materials were used as the primary basis. Selected regulatory values were cross-checked against current U.S. sources (OSHA, EPA, FEMA/USFA, CDC/NIOSH, FMCSA and PHMSA). Code-specific values are identified where the recalled wording is not sufficient for a universal answer.

1. System Safety: Inductive vs Deductive

Q1. A safety engineer begins with a pump seal failure and evaluates the possible effects on the system. Which technique BEST matches this approach?

A. Fault Tree Analysis (FTA)

B. Failure Modes and Effects Analysis (FMEA)

C. Management Oversight and Risk Tree (MORT)

D. Bow-tie analysis

Recall note: Inductive analysis starts with a component failure or initiating event and asks what consequences may follow. FMEA and ETA are commonly treated as inductive/forward techniques. FTA starts with an undesired top event and works backward, so it is deductive. A recalled exam item that offers both ETA and FMEA as 'inductive' can be ambiguous; use the scenario wording to distinguish them.

Q2. Which technique starts with an undesired top event and works backward to identify combinations of causes?

A. Event Tree Analysis (ETA)

B. FMEA

C. Fault Tree Analysis (FTA)

D. HAZOP

Recall note: Inductive analysis starts with a component failure or initiating event and asks what consequences may follow. FMEA and ETA are commonly treated as inductive/forward techniques. FTA starts with an undesired top event and works backward, so it is deductive. A recalled exam item that offers both ETA and FMEA as 'inductive' can be ambiguous; use the scenario wording to distinguish them.

2. FTA: Cut Sets and Logic Gates

Q3. Minimum cut sets are most directly associated with which analysis?

A. ETA

B. FMEA

C. FTA

D. HAZOP

Recall note: FTA uses logic gates to connect causes to a top event. An AND gate requires all input events; an OR gate requires any one input. A minimal cut set is the smallest combination of basic events sufficient to cause the top event.

Q4. In an FTA, two independent failures must BOTH occur for the top event. Which gate represents this relationship?

A. OR gate

B. AND gate

C. Transfer gate

D. Inhibit-free gate

Recall note: FTA uses logic gates to connect causes to a top event. An AND gate requires all input events; an OR gate requires any one input. A minimal cut set is the smallest combination of basic events sufficient to cause the top event.

3. HAZOP and Guide Words

Q5. A team uses MORE, LESS, NO and REVERSE to challenge process parameters. Which technique is being used?

A. FTA

B. FMEA

C. HAZOP

D. ETA

Recall note: HAZOP is a structured, mainly qualitative process-hazard study. It combines process parameters with guide words to identify deviations from design intent. Typical guide words include NO/NOT, MORE, LESS, REVERSE, AS WELL AS, PART OF, OTHER THAN, EARLY and LATE. 'Overflow' may be the deviation produced by MORE level/flow; the key clue is guide words + process parameters.

Q6. The phrase 'MORE FLOW' during a process review represents:

A. A HAZOP deviation

B. An FTA cut set

C. An FMEA detection rating

D. A Pareto category

Recall note: HAZOP is a structured, mainly qualitative process-hazard study. It combines process parameters with guide words to identify deviations from design intent. Typical guide words include NO/NOT, MORE, LESS, REVERSE, AS WELL AS, PART OF, OTHER THAN, EARLY and LATE. 'Overflow' may be the deviation produced by MORE level/flow; the key clue is guide words + process parameters.

4. FMEA and RPN

Q7. Which technique systematically evaluates failure modes, their effects, and often severity, occurrence and detection?

A. FMEA
B. FTA
C. HAZOP only
D. ETA

Recall note: FMEA is a forward, bottom-up technique that asks how components/process steps can fail and what effects result. Traditional FMEA often prioritizes failure modes using RPN = Severity x Occurrence x Detection. Higher RPN generally means greater priority for action, but RPN should not replace engineering judgment.

Q8. In a traditional FMEA, RPN is commonly calculated as:

A. Severity + Occurrence + Detection
B. Severity x Occurrence x Detection
C. Probability x Cost
D. Frequency / Severity

Recall note: FMEA is a forward, bottom-up technique that asks how components/process steps can fail and what effects result. Traditional FMEA often prioritizes failure modes using RPN = Severity x Occurrence x Detection. Higher RPN generally means greater priority for action, but RPN should not replace engineering judgment.

5. Ladder Capacity and 4:1 Setup

Q9. A portable construction ladder must generally be capable of supporting at least how many times its maximum intended load?

A. 1 time
B. 2 times
C. 3.5 times
D. 4 times

Recall note: For OSHA construction portable ladders, self-supporting and non-self-supporting ladders generally must support at least 4 times the maximum intended load; the extra-heavy-duty Type IA metal/plastic exception is 3.3 times. The 4:1 setup rule means the base is about 1 ft out for every 4 ft of working ladder length/vertical support relationship. Do not confuse design strength with the manufacturer's rated user load.

Q10. A non-self-supporting ladder reaches a support point about 20 ft above the base. Approximately how far should the foot be from the support using the 4:1 rule?

A. 2 ft
B. 5 ft
C. 10 ft
D. 20 ft

Recall note: For OSHA construction portable ladders, self-supporting and non-self-supporting ladders generally must support at least 4 times the maximum intended load; the extra-heavy-duty Type IA metal/plastic exception is 3.3 times. The 4:1 setup rule means the base is about 1 ft out for every 4 ft of working ladder length/vertical support relationship. Do not confuse design strength with the manufacturer's rated user load.

6. Universal Waste

Q11. Which item is a federal universal waste category?

A. Discarded fluorescent lamp
B. COVID-19 medical waste
C. Food waste
D. Untreated sewage

Recall note: Federal EPA universal waste categories are batteries, certain pesticides, mercury-containing equipment, lamps, and aerosol cans. COVID-19 waste is not itself a federal universal-waste category. States may add categories.

Q12. Which list contains only federal universal-waste categories?

A. Batteries, lamps, mercury-containing equipment

B. Used oil, sewage, lamps

C. Medical waste, lamps, soil

D. Asbestos, batteries, food waste

Recall note: Federal EPA universal waste categories are batteries, certain pesticides, mercury-containing equipment, lamps, and aerosol cans. COVID-19 waste is not itself a federal universal-waste category. States may add categories.

7. ICS Safety Officer

Q13. Who in the Incident Command System monitors safety conditions and develops measures to protect responders?

A. Incident Commander

B. Safety Officer

C. Liaison Officer

D. Planning Section Chief

Recall note: In ICS, the Incident Commander has overall responsibility. The Safety Officer monitors safety conditions, assesses hazards and develops measures to protect assigned personnel; the role may stop/prevent unsafe acts. Liaison coordinates assisting agencies; PIO handles public information.

Q14. Who has overall responsibility for incident objectives, strategy and priorities?

A. Safety Officer

B. Logistics Section Chief

C. Incident Commander

D. Public Information Officer

Recall note: In ICS, the Incident Commander has overall responsibility. The Safety Officer monitors safety conditions, assesses hazards and develops measures to protect assigned personnel; the role may stop/prevent unsafe acts. Liaison coordinates assisting agencies; PIO handles public information.

8. Respiratory Protection and APF

Q15. Which option generally provides the highest assigned protection factor?

A. Half-mask APR

B. Full-face APR

C. Pressure-demand SCBA

D. Filtering facepiece

Recall note: OSHA APFs depend on respirator type and mode. Typical values: half-mask APR 10; full-face APR 50; pressure-demand/positive-pressure full-face SCBA 10,000. Therefore, among SCBA, full-face APR, half-mask APR and cartridge APR, a positive-pressure SCBA provides the highest protection. For IDLH, use appropriate atmosphere-supplying respiratory protection, not an APR.

Q16. Under OSHA APF tables, a full-face air-purifying respirator typically has an APF of:

A. 10

B. 25

C. 50

D. 10,000

Recall note: OSHA APFs depend on respirator type and mode. Typical values: half-mask APR 10; full-face APR 50; pressure-demand/positive-pressure full-face SCBA 10,000. Therefore, among SCBA, full-face APR, half-mask APR and cartridge APR, a positive-pressure SCBA provides the highest protection. For IDLH, use appropriate atmosphere-supplying respiratory protection, not an APR.

9. Transformational Leadership

Q17. A manager introduces a new safety vision, inspires employees and motivates them to challenge old practices. Which style is MOST evident?

A. Transactional

B. Transformational

C. Delegative

D. Autocratic

Recall note: Transformational leaders create a compelling vision, inspire people, encourage improvement and change, and motivate followers beyond routine transactions. Transactional leadership relies more on rewards/corrections; delegative leadership gives substantial decision freedom; autocratic leadership directs decisions.

Q18. Which leadership style is most closely associated with rewards for meeting defined performance expectations?

A. Transformational

B. Transactional

C. Servant

D. Laissez-faire

Recall note: Transformational leaders create a compelling vision, inspire people, encourage improvement and change, and motivate followers beyond routine transactions. Transactional leadership relies more on rewards/corrections; delegative leadership gives substantial decision freedom; autocratic leadership directs decisions.

10. ESG

Q19. What does ESG stand for?

A. Environmental, Safety, Government

B. Environmental, Social, Governance

C. Energy, Sustainability, Growth

D. Environment, Standards, Guidance

Recall note: ESG means Environmental, Social and Governance. It is a framework used to consider sustainability and organizational impacts/risks; it is not simply another name for EPA regulation. Environmental covers matters such as emissions/resource use; Social covers people/community; Governance covers oversight, ethics and accountability.

Q20. Board oversight, ethics and accountability fall primarily under which ESG pillar?

A. Environmental

B. Social

C. Governance

D. Engineering

Recall note: ESG means Environmental, Social and Governance. It is a framework used to consider sustainability and organizational impacts/risks; it is not simply another name for EPA regulation. Environmental covers matters such as emissions/resource use; Social covers people/community; Governance covers oversight, ethics and accountability.

11. Forklift Fork Height / Travel

Q21. While traveling with a load, where should forklift forks/load normally be carried?

A. As high as visibility permits

B. About 4-6 inches above the ground

C. At operator shoulder height

D. Touching the ground

Recall note: When traveling, a forklift load/forks should be kept as low as practical. OSHA's forklift eTool recommends carrying the load at the lowest possible position, about 4-6 inches from the ground, with the mast tilted back as appropriate. Never travel with the load unnecessarily elevated.

Q22. Why should a forklift not travel with the load elevated?

A. It reduces tire pressure

B. It raises/shifts the combined center of gravity and reduces stability

C. It increases battery voltage

D. It improves braking too much

Recall note: When traveling, a forklift load/forks should be kept as low as practical. OSHA's forklift eTool recommends carrying the load at the lowest possible position, about 4-6 inches from the ground, with the mast tilted back as appropriate. Never travel with the load unnecessarily elevated.

12. Business Continuity Planning

Q23. Who should participate in development of a business continuity plan?

A. Only the HSE department

B. Only the company director

C. A cross-functional team representing critical business functions

D. Only Finance and HR

Recall note: BCP aims to maintain or restore critical business functions during and after disruption. It is cross-functional: leadership plus operations, HR, IT, finance, HSE/security and other critical functions as applicable. HR focuses on people continuity: essential staffing, contact information, welfare, succession and workforce arrangements.

Q24. During continuity planning, HR should MOST importantly ensure that:

A. Every employee reports to the normal workplace regardless of conditions

B. Essential personnel and staffing needs are identified and addressed

C. All annual leave is cancelled permanently

D. Only managers receive emergency contact information

Recall note: BCP aims to maintain or restore critical business functions during and after disruption. It is cross-functional: leadership plus operations, HR, IT, finance, HSE/security and other critical functions as applicable. HR focuses on people continuity: essential staffing, contact information, welfare, succession and workforce arrangements.

13. Fire Classes and Electrical Equipment

Q25. A fire involving energized electrical equipment is classified as:

A. Class A

B. Class B

C. Class C

D. Class D

Recall note: Class A = ordinary combustibles; B = flammable liquids/gases; C = energized electrical equipment; D = combustible metals; K = cooking oils/fats. For energized electrical equipment, use an agent suitable for Class C/nonconductive use. Once de-energized, classify by the underlying fuel.

Q26. After electrical equipment is de-energized, extinguisher selection should primarily be based on:

A. The underlying burning material

B. The voltage that was present

C. The wire color

D. The circuit-breaker rating

Recall note: Class A = ordinary combustibles; B = flammable liquids/gases; C = energized electrical equipment; D = combustible metals; K = cooking oils/fats. For energized electrical equipment, use an agent suitable for Class C/nonconductive use. Once de-energized, classify by the underlying fuel.

14. Water Solubility, Specific Gravity and Float/Sink

Q27. A liquid forms a separate layer on water and may float or sink depending on its density. It is BEST described as:

A. Water-soluble liquid

B. Water-soluble solid

C. Water-insoluble liquid

D. Water-insoluble solid

Recall note: Solubility tells whether a substance mixes/dissolves in water. Specific gravity (SG) predicts float/sink for a separate insoluble phase: SG < 1 tends to float; SG > 1 tends to sink. If a liquid remains as a separate layer and may float or sink, 'water-insoluble liquid' is the key classification.

Q28. An insoluble liquid with specific gravity 0.8 will generally:

A. Sink

B. Float

C. Dissolve completely

D. Become a solid

Recall note: Solubility tells whether a substance mixes/dissolves in water. Specific gravity (SG) predicts float/sink for a separate insoluble phase: SG < 1 tends to float; SG > 1 tends to sink. If a liquid remains as a separate layer and may float or sink, 'water-insoluble liquid' is the key classification.

15. Training: Bee-Sting First Aid Demonstration

Q29. A trainee is asked to demonstrate first aid for a simulated bee sting during landscape training. Which delivery method BEST fits?

A. Lecture

B. Simulation

C. Case study only

D. Reading assignment

Recall note: When a trainee practices a response to a simulated event, the delivery method is simulation. The learning domain is psychomotor/kinesthetic because the trainee physically demonstrates a skill. OJT is supervised learning while performing real job tasks in the work setting.

Q30. The trainee physically demonstrates the first-aid steps. Which learning domain is emphasized?

A. Cognitive only

B. Affective only

C. Psychomotor/kinesthetic

D. Financial

Recall note: When a trainee practices a response to a simulated event, the delivery method is simulation. The learning domain is psychomotor/kinesthetic because the trainee physically demonstrates a skill. OJT is supervised learning while performing real job tasks in the work setting.

16. Epidemiology: Risk, Mortality and Case Fatality

Q31. A pediatric-ward study compares infection risk in exposed children with infection risk in unexposed children. Which measure is MOST appropriate?

A. Relative risk

B. Case fatality proportion

C. Crude birth rate

D. Maturity ratio

Recall note: Risk ratio (relative risk) compares disease risk in an exposed group with risk in an unexposed group: RR = risk exposed / risk unexposed. Mortality rate uses deaths in a population over a period. Case-fatality proportion compares deaths from a disease with diagnosed cases of that disease. Do not interchange these measures.

Q32. During an outbreak, 56 of 400 diagnosed patients die. The case-fatality proportion is:

A. 7%

B. 14%

C. 56%

D. 71.4%

Recall note: Risk ratio (relative risk) compares disease risk in an exposed group with risk in an unexposed group: RR = risk exposed / risk unexposed. Mortality rate uses deaths in a population over a period. Case-fatality proportion compares deaths from a disease with diagnosed cases of that disease. Do not interchange these measures.

17. ISO 14001 Employee Awareness

Q33. Under an ISO 14001 EMS, which item should employees be aware of?

A. Environmental policy and relevant environmental aspects

B. Only annual profit

C. Only tax records

D. Only emergency phone numbers

Recall note: ISO 14001 is an environmental management system standard. Employees should be aware of the environmental policy, significant environmental aspects/impacts relevant to their work, their contribution to EMS effectiveness, and implications of not conforming. 'Aspect' is an element of activities/products/services that can interact with the environment; 'impact' is the resulting environmental change.

Q34. A discharge from an activity is an environmental aspect; contamination of a stream would be the:

A. Policy

B. Impact

C. Audit criterion

D. Objective owner

Recall note: ISO 14001 is an environmental management system standard. Employees should be aware of the environmental policy, significant environmental aspects/impacts relevant to their work, their contribution to EMS effectiveness, and implications of not conforming. 'Aspect' is an element of activities/products/services that can interact with the environment; 'impact' is the resulting environmental change.

18. Brainstorming

Q35. A group freely proposes many solutions to a safety problem before evaluating them. Which method is being used?

A. Role play

B. Brainstorming

C. Lecture

D. Job rotation

Recall note: Brainstorming is a group idea-generation method. Participants generate many ideas before judging them; evaluation/prioritization follows later. Role play involves acting assigned roles; nominal group technique is more structured and often includes individual idea generation and ranking.

Q36. During brainstorming, the facilitator should initially:

A. Criticize weak ideas immediately

B. Encourage idea generation and defer judgment

C. Allow only managers to speak

D. Rank each idea before the next is offered

Recall note: Brainstorming is a group idea-generation method. Participants generate many ideas before judging them; evaluation/prioritization follows later. Role play involves acting assigned roles; nominal group technique is more structured and often includes individual idea generation and ranking.

19. GHS: Signal Word vs Hazard Statement

Q37. Which GHS label element indicates the relative severity of the hazard?

A. Signal word

B. Hazard statement

C. Product identifier

D. Supplier address

Recall note: Signal word indicates the relative level of hazard severity: DANGER is used for more severe hazard categories and WARNING for less severe categories where a signal word is assigned. A hazard statement describes the nature of the hazard and, where appropriate, the degree of hazard. Precautionary statements tell what to do to prevent/respond.

Q38. Which is an example of a hazard statement?

A. DANGER

B. Wear protective gloves

C. Highly flammable liquid and vapor

D. ABC Chemical Company

Recall note: Signal word indicates the relative level of hazard severity: DANGER is used for more severe hazard categories and WARNING for less severe categories where a signal word is assigned. A hazard statement describes the nature of the hazard and, where appropriate, the degree of hazard. Precautionary statements tell what to do to prevent/respond.

20. Driver Alcohol/Drug Testing

Q39. A CDL driver is selected through the employer's random testing program while performing safety-sensitive work. What type of test is this?

A. Random

B. Post-accident

C. Return-to-duty

D. Follow-up

Recall note: FMCSA/DOT testing circumstances include random, reasonable suspicion, post-accident, return-to-duty and follow-up; controlled-substance testing also includes pre-employment. Alcohol random testing must occur while the driver is performing, just before, or just after safety-sensitive functions. Reasonable suspicion is based on trained observations, not unexplained absence alone.

Q40. A supervisor observes signs consistent with alcohol impairment in a driver. Which test basis is MOST appropriate?

A. Reasonable suspicion

B. Routine medical exam

C. Tax verification

D. Vehicle inspection

Recall note: FMCSA/DOT testing circumstances include random, reasonable suspicion, post-accident, return-to-duty and follow-up; controlled-substance testing also includes pre-employment. Alcohol random testing must occur while the driver is performing, just before, or just after safety-sensitive functions. Reasonable suspicion is based on trained observations, not unexplained absence alone.

21. Pre-employment Screening and Driver Selection

Q41. Which is the MOST job-relevant pre-screening record when selecting a professional driver?

A. Tax record

B. Motor vehicle/driving record

C. Personal social-media password

D. Unrelated family medical record

Recall note: Pre-employment screening may include lawful, job-related background, qualification, reference and driving-record checks. For fleet safety, the motor vehicle/driving record is especially relevant because it shows prior driving history. Medical/disability inquiries in the U.S. are legally restricted before a conditional job offer.

Q42. A driver has passed interview and general background checks. What should be reviewed to identify prior risk-taking driving behavior?

A. Driving record

B. Favorite vehicle model

C. Personal tax return

D. Home utility bill

Recall note: Pre-employment screening may include lawful, job-related background, qualification, reference and driving-record checks. For fleet safety, the motor vehicle/driving record is especially relevant because it shows prior driving history. Medical/disability inquiries in the U.S. are legally restricted before a conditional job offer.

22. Remedial Training

Q43. A driver has repeated preventable incidents caused by poor following distance. What training is MOST appropriate?

A. Generic induction only

B. Remedial training focused on the identified deficiency

C. Payroll training

D. No training because the driver is experienced

Recall note: Remedial training is targeted corrective training used when performance, incidents or observed deficiencies show a specific skill/behavior gap. It should address the causal deficiency, not merely repeat generic orientation.

Q44. The purpose of remedial training is to:

A. Correct an identified performance deficiency

B. Replace all engineering controls

C. Punish the employee

D. Avoid incident investigation

Recall note: Remedial training is targeted corrective training used when performance, incidents or observed deficiencies show a specific skill/behavior gap. It should address the causal deficiency, not merely repeat generic orientation.

23. Records and Backup / Secondary Copies

Q45. Which method BEST protects critical business records from loss at the primary site?

A. One paper copy beside the original

B. Controlled electronic backup stored separately/offsite

C. Employee's personal folder

D. Emailing an uncontrolled copy to anyone

Recall note: Critical records should have controlled, retrievable backups. A robust continuity approach uses secure redundant electronic copies and, where appropriate, offsite/geographically separated storage. Personal uncontrolled folders are not a sound records-management method.

Q46. The main purpose of a secondary backup is to provide:

A. Redundancy and recoverability

B. A second uncontrolled master

C. A reason to destroy the original

D. A substitute for access control

Recall note: Critical records should have controlled, retrievable backups. A robust continuity approach uses secure redundant electronic copies and, where appropriate, offsite/geographically separated storage. Personal uncontrolled folders are not a sound records-management method.

24. Management of Change (MOC)

Q47. A new solvent is proposed for an existing process. Which process should evaluate the change before use?

A. MOC

B. Pareto analysis

C. Payroll audit

D. Incident rate calculation

Recall note: MOC is a formal process for evaluating changes before implementation so new hazards are not unintentionally introduced. Typical elements include technical basis, safety/health impact, procedure changes, duration, authorization,

training/communication and updates to affected documentation.

Q48. What is the primary safety purpose of MOC?

A. Increase production quantity regardless of risk
B. Identify and control risks introduced by change
C. Reduce documentation
D. Transfer all risk to workers

Recall note: MOC is a formal process for evaluating changes before implementation so new hazards are not unintentionally introduced. Typical elements include technical basis, safety/health impact, procedure changes, duration, authorization, training/communication and updates to affected documentation.

25. CO2, Foam and Class B Fire

Q49. AFFF is primarily associated with control of which fire class?

A. Class A only
B. Class B
C. Class D
D. Class K only

Recall note: AFFF is principally associated with Class B flammable-liquid fires because it forms an aqueous film/blanket over hydrocarbon fuel. CO2 is suitable for Class B and Class C hazards; it leaves no residue but has limited cooling and can create asphyxiation hazards in confined spaces. If an exam asks for the common class shared by CO2 and AFFF, Class B is the best answer.

Q50. If a question asks for the fire class commonly suitable for BOTH CO2 and AFFF, the BEST answer is:

A. Class A
B. Class B
C. Class D
D. Class K

Recall note: AFFF is principally associated with Class B flammable-liquid fires because it forms an aqueous film/blanket over hydrocarbon fuel. CO2 is suitable for Class B and Class C hazards; it leaves no residue but has limited cooling and can create asphyxiation hazards in confined spaces. If an exam asks for the common class shared by CO2 and AFFF, Class B is the best answer.

26. NIOSH RWL and Lifting Index

Q51. In ergonomics, RWL means:

A. Recommended Weight Load
B. Recommended Weight Limit
C. Required Work Level
D. Rated Working Load

Recall note: RWL means Recommended Weight Limit in the Revised NIOSH Lifting Equation. The load constant is 51 lb (23 kg) under ideal conditions, then multipliers reduce the RWL as conditions worsen. LI = actual load / RWL. LI above 1 indicates increasing lifting-related risk. 51 lb is not a universal legal maximum lift.

Q52. A load is 40 lb and the calculated RWL is 20 lb. What is the lifting index?

A. 0.5
B. 1.0
C. 2.0
D. 20

Recall note: RWL means Recommended Weight Limit in the Revised NIOSH Lifting Equation. The load constant is 51 lb (23 kg) under ideal conditions, then multipliers reduce the RWL as conditions worsen. LI = actual load / RWL. LI above 1 indicates increasing lifting-related risk. 51 lb is not a universal legal maximum lift.

27. Combined Noise and Hearing Protection

Q53. Two independent machines produce 75 dBA and 85 dBA at the worker. The combined level is closest to:

A. 85 dBA
B. 85.4 dBA
C. 95 dBA
D. 160 dBA

Recall note: Decibels are logarithmic, so 75 dBA + 85 dBA is about 85.4 dBA, not 160 dBA. Under OSHA, 85 dBA 8-hour TWA is the hearing-conservation action level and 90 dBA is the 8-hour PEL; hearing protectors are not automatically mandatory for

every worker merely because a measurement is near 85 dBA. ACGIH/NIOSH use more protective criteria, so always read which standard and duration the question specifies.

Q54. In a recalled exam scenario, combined exposure is about 85 dBA and no additional condition requiring hearing protectors is stated. Which answer is MOST likely intended?

A. Double hearing protection
B. Earmuffs only
C. PPE not automatically required based on that value alone
D. SCBA

Recall note: Decibels are logarithmic, so 75 dBA + 85 dBA is about 85.4 dBA, not 160 dBA. Under OSHA, 85 dBA 8-hour TWA is the hearing-conservation action level and 90 dBA is the 8-hour PEL; hearing protectors are not automatically mandatory for every worker merely because a measurement is near 85 dBA. ACGIH/NIOSH use more protective criteria, so always read which standard and duration the question specifies.

28. Herzberg Two-Factor Theory

Q55. Which is a Herzberg motivator?

A. Recognition
B. Salary
C. Supervision
D. Company policy

Recall note: Herzberg separates motivators from hygiene factors. Motivators include achievement, recognition, responsibility, advancement, growth and the work itself. Hygiene factors include salary/pay, supervision, company policy, working conditions, relationships and job security. Improving hygiene can reduce dissatisfaction; motivators support satisfaction/motivation.

Q56. Which is a Herzberg hygiene factor?

A. Achievement
B. Recognition
C. Salary
D. Advancement

Recall note: Herzberg separates motivators from hygiene factors. Motivators include achievement, recognition, responsibility, advancement, growth and the work itself. Hygiene factors include salary/pay, supervision, company policy, working conditions, relationships and job security. Improving hygiene can reduce dissatisfaction; motivators support satisfaction/motivation.

29. 6M, Fishbone, 5-Why and Pareto

Q57. The 6M method is commonly used with which root-cause tool?

A. Fishbone/Ishikawa diagram
B. FTA only
C. ETA only
D. Gantt chart

Recall note: The 6M categories are commonly Man/People, Machine, Method, Material, Measurement and Mother Nature/Environment. They structure a Fishbone/Ishikawa cause-and-effect analysis. 5-Why drills down a causal chain. Pareto prioritizes the vital few causes, often remembered as roughly 80% of effects from 20% of causes.

Q58. After identifying many causes, which tool helps prioritize the few causes producing most of the problems?

A. Pareto analysis
B. Role play
C. HAZOP guide words
D. SDS

Recall note: The 6M categories are commonly Man/People, Machine, Method, Material, Measurement and Mother Nature/Environment. They structure a Fishbone/Ishikawa cause-and-effect analysis. 5-Why drills down a causal chain. Pareto prioritizes the vital few causes, often remembered as roughly 80% of effects from 20% of causes.

30. Infectious Substance Shipping: UN 2814 / UN 2900

Q59. A Category A infectious substance can cause disease in humans. Which UN number applies?

A. UN 2814
B. UN 2900
C. UN 1203
D. UN 1950

Recall note: For Category A infectious substances, UN 2814 is 'Infectious substance, affecting humans'; UN 2900 is 'Infectious substance, affecting animals only.' If the agent can affect humans (even if it may also affect animals), UN 2814 is the key classification. Shipping papers require the proper basic description and applicable technical-name requirements.

Q60. UN 2900 is used for:

A. Infectious substances affecting humans
B. Infectious substances affecting animals only
C. Flammable liquids
D. Aerosol cans

Recall note: For Category A infectious substances, UN 2814 is 'Infectious substance, affecting humans'; UN 2900 is 'Infectious substance, affecting animals only.' If the agent can affect humans (even if it may also affect animals), UN 2814 is the key classification. Shipping papers require the proper basic description and applicable technical-name requirements.

31. Prevention through Design (PtD)

Q61. What is the fundamental advantage of Prevention through Design?

A. It postpones controls until operations begin
B. It eliminates/reduces hazards early and can avoid later retrofitting
C. It relies mainly on PPE
D. It removes the need for risk assessment

Recall note: PtD integrates hazard elimination and risk reduction into design decisions early in the lifecycle. The main advantage is avoiding hazards and costly retrofits later. Hazard identification, analysis and evaluation are assessment activities; elimination/substitution/engineering are control choices.

Q62. During PtD, what should occur before selecting controls?

A. Hazard identification and risk analysis/evaluation
B. Purchase PPE only
C. Wait for an injury
D. Ignore foreseeable maintenance tasks

Recall note: PtD integrates hazard elimination and risk reduction into design decisions early in the lifecycle. The main advantage is avoiding hazards and costly retrofits later. Hazard identification, analysis and evaluation are assessment activities; elimination/substitution/engineering are control choices.

32. Means of Egress

Q63. What is the correct sequence of the three parts of an exit route?

A. Exit - exit access - exit discharge
B. Exit access - exit - exit discharge
C. Exit discharge - exit - exit access
D. Access - discharge - exit

Recall note: A means/exit route has three parts in order: exit access -> exit -> exit discharge. Exit access leads to the protected exit; the exit leads toward the discharge; exit discharge leads outside or to a safe public/open area. Exit routes must be adequately lighted and unobstructed.

Q64. Which part leads directly outside or to a safe open/public way?

A. Exit access
B. Exit discharge
C. Work area
D. Dead-end corridor

Recall note: A means/exit route has three parts in order: exit access -> exit -> exit discharge. Exit access leads to the protected exit; the exit leads toward the discharge; exit discharge leads outside or to a safe public/open area. Exit routes must be adequately lighted and unobstructed.

33. Emergency Lighting

Q65. A CSP-style life-safety question asks the commonly required minimum duration for emergency lighting after normal power failure. Which value is typically expected?

A. 30 minutes
B. 45 minutes
C. 60 minutes
D. 90 minutes

Recall note: OSHA requires exit routes to be adequately lighted and emergency safeguards kept operable. The commonly tested 90-minute emergency-lighting duration comes from life-safety/building-code practice, not a general OSHA 1910.37 sentence specifying 90 minutes. Treat 90 minutes as a code-based exam number when that context is clear.

Q66. Under OSHA exit-route rules, the core requirement is that exit routes be:

A. Dark to preserve backup power
B. Adequately lighted and safeguards maintained
C. Used for storage
D. Locked during normal work

Recall note: OSHA requires exit routes to be adequately lighted and emergency safeguards kept operable. The commonly tested 90-minute emergency-lighting duration comes from life-safety/building-code practice, not a general OSHA 1910.37 sentence specifying 90 minutes. Treat 90 minutes as a code-based exam number when that context is clear.

34. Hydrostatic Test - Read the Equipment Context

Q67. A recalled CSP source specifically asks pressure-vessel hydrostatic test pressure and gives 150% of MOP as the intended value. Which answer matches that source?

A. 100%
B. 125%
C. 150%
D. 200%

Recall note: Hydrostatic test pressure is code/equipment-specific. A recalled source states 150% of maximum operating pressure for a pressure-vessel question. Do not automatically apply 150% to every piping system; piping codes can use different formulas/ratios. CSP questions often test whether you notice the equipment/code context.

Q68. What is the safest exam approach when a question asks hydrotest pressure for 'piping' without identifying the governing code?

A. Assume every system is 150%
B. Recognize that test pressure is code/system-specific and use the stated governing requirement
C. Always use 200%
D. Use the fire-load formula

Recall note: Hydrostatic test pressure is code/equipment-specific. A recalled source states 150% of maximum operating pressure for a pressure-vessel question. Do not automatically apply 150% to every piping system; piping codes can use different formulas/ratios. CSP questions often test whether you notice the equipment/code context.

35. Wire Rope Diameter Measurement

Q69. How should wire-rope diameter be measured?

A. Valley to valley
B. Crown to crown across opposite strands
C. Around one wire only
D. Across the hook throat

Recall note: Wire-rope diameter is measured across the crowns of opposite strands, not across valleys. Inspection/measurement should be representative; exact required locations depend on the applicable rope/equipment standard and manufacturer. The recalled 'minimum points' question is incomplete, so memorize the correct crown-to-crown method rather than inventing a universal number.

Q70. A question asks for a universal minimum number of measurement points but gives no rope/equipment standard. What is the BEST interpretation?

A. There is one universal number for every wire rope
B. The requirement is standard/application-specific; use representative measurements and the governing criterion
C. Never measure diameter
D. Measure only after failure

Recall note: Wire-rope diameter is measured across the crowns of opposite strands, not across valleys. Inspection/measurement should be representative; exact required locations depend on the applicable rope/equipment standard and manufacturer. The recalled 'minimum points' question is incomplete, so memorize the correct crown-to-crown method rather than inventing a universal number.

36. Tedlar Bag Sampling

Q71. A Tedlar bag is MOST commonly used to collect:

A. Gas/vapor air samples

B. Settled dust by wiping

C. Noise samples

D. Radiation dose

Recall note: A Tedlar bag is used to collect a grab/integrated sample of gases or vapors for later analysis. It is not a particulate filter. Bags are useful when the contaminant is compatible with the bag material and sample stability is adequate.

Q72. A single air sample captured in a bag at a particular time is BEST described as:

A. Grab sampling

B. Dosimetry

C. Surface wipe sampling

D. Biological monitoring

Recall note: A Tedlar bag is used to collect a grab/integrated sample of gases or vapors for later analysis. It is not a particulate filter. Bags are useful when the contaminant is compatible with the bag material and sample stability is adequate.

37. Flame Spread Index and Fire Load

Q73. Which term describes the relative rate at which flame spreads over a material's surface in a standard test?

A. Fire load

B. Flame Spread Index

C. Flash point

D. LEL

Recall note: Flame Spread Index (FSI) describes relative surface flame-spread behavior of materials in a standardized test; lower values indicate slower surface flame spread. Fire load is the total potential heat energy from combustibles, often expressed per unit floor area: sum(mass x heat of combustion) / floor area.

Q74. Fire-load density is MOST directly calculated from:

A. Total potential heat from combustibles divided by floor area

B. Floor area divided by number of exits

C. Temperature x humidity

D. Noise dose x time

Recall note: Flame Spread Index (FSI) describes relative surface flame-spread behavior of materials in a standardized test; lower values indicate slower surface flame spread. Fire load is the total potential heat energy from combustibles, often expressed per unit floor area: sum(mass x heat of combustion) / floor area.

38. Head, Heart and Hands

Q75. In the Head-Heart-Hands model, 'Hands' corresponds most closely to:

A. Cognitive learning

B. Affective learning

C. Psychomotor learning

D. Financial learning

Recall note: A common training memory model maps Head = cognitive knowledge/thinking, Heart = affective attitudes/values, and Hands = psychomotor skills/doing. A demonstration primarily assesses psychomotor performance.

Q76. Changing a worker's safety values and attitudes primarily targets which domain?

A. Cognitive

B. Affective

C. Psychomotor

D. Mechanical

Recall note: A common training memory model maps Head = cognitive knowledge/thinking, Heart = affective attitudes/values, and Hands = psychomotor skills/doing. A demonstration primarily assesses psychomotor performance.

39. Leading and Lagging Indicators

Q77. Routine housekeeping inspection completion is BEST classified as:

A. Leading indicator

B. Lagging indicator

C. Financial ratio

D. Exposure limit

Recall note: Leading indicators are proactive measures of activities/conditions intended to prevent harm: inspections, training completion, corrective-action closure, safety meetings and preventive maintenance. Lagging indicators measure outcomes

after events: injuries, lost days, TRIR, DART, property damage. Near misses can be used proactively, but counts of past near misses are still event/outcome data; wording matters.

Q78. Which is a lagging indicator?

A. Percentage of corrective actions closed

B. Number of preventive inspections

C. TRIR

D. Safety observations completed

Recall note: Leading indicators are proactive measures of activities/conditions intended to prevent harm: inspections, training completion, corrective-action closure, safety meetings and preventive maintenance. Lagging indicators measure outcomes after events: injuries, lost days, TRIR, DART, property damage. Near misses can be used proactively, but counts of past near misses are still event/outcome data; wording matters.

40. Shoveling and Manual Handling

Q79. Which change BEST reduces ergonomic stress during repetitive shoveling?

A. Throw material farther

B. Reduce shovel load and throwing distance

C. Twist rapidly at the waist

D. Use the heaviest shovel available

Recall note: Manual shoveling risk rises with heavy loads, long reaches, twisting and throwing material far. Reduce load weight, keep work close, use appropriate shovel size/handle, avoid twisting, alternate tasks and use mechanical assistance when practical. 'Double insulated shovel' is not a general ergonomic control.

Q80. The primary injury concern from repetitive manual shoveling is:

A. Musculoskeletal disorder

B. Radiation sickness

C. Hearing loss only

D. Electrical arc flash only

Recall note: Manual shoveling risk rises with heavy loads, long reaches, twisting and throwing material far. Reduce load weight, keep work close, use appropriate shovel size/handle, avoid twisting, alternate tasks and use mechanical assistance when practical. 'Double insulated shovel' is not a general ergonomic control.

41. Ill-Fitting Gloves

Q81. What is a key hazard of loose, ill-fitting gloves near rotating equipment?

A. Improved dexterity

B. Entanglement

C. Lower noise exposure

D. Reduced friction only

Recall note: Gloves must fit the task and worker. Oversized/loose gloves reduce dexterity and can be caught in moving machinery; overly tight gloves can restrict movement/circulation and may worsen fatigue, especially with vibration. Chemical glove selection also depends on permeation, degradation and breakthrough time.

Q82. When selecting chemical-resistant gloves, which property is especially important?

A. Permeation/breakthrough performance

B. Glove color only

C. Employee shoe size

D. Room lighting

Recall note: Gloves must fit the task and worker. Oversized/loose gloves reduce dexterity and can be caught in moving machinery; overly tight gloves can restrict movement/circulation and may worsen fatigue, especially with vibration. Chemical glove selection also depends on permeation, degradation and breakthrough time.

42. Delegation vs Centralization

Q83. Management gives frontline supervisors authority to take defined corrective action for misconduct. This is:

A. Centralization

B. Delegation

C. Abdication

D. Subrogation

Recall note: Delegation means assigning authority and responsibility downward while the manager retains accountability for overall results. Centralization keeps decision authority concentrated at higher levels. Giving a frontline supervisor authority to

address routine misconduct is delegation.

Q84. Delegation transfers:

A. Authority for specified tasks while higher management retains accountability
B. All organizational accountability permanently
C. Ownership of the company
D. Legal standards

Recall note: Delegation means assigning authority and responsibility downward while the manager retains accountability for overall results. Centralization keeps decision authority concentrated at higher levels. Giving a frontline supervisor authority to address routine misconduct is delegation.

43. Maximum Manual Lift Weight

Q85. What is the NIOSH Revised Lifting Equation load constant under ideal conditions?

A. 23 lb
B. 35 lb
C. 51 lb
D. 100 lb

Recall note: There is no single OSHA general-industry legal maximum weight that every worker may lift. The NIOSH RNLE uses a 51-lb load constant under ideal conditions and adjusts it with task multipliers to calculate RWL. The correct exam response depends on whether the question asks for the NIOSH load constant or a legal maximum.

Q86. Which statement is correct about a universal legal maximum manual-lifting weight under OSHA?

A. It is always 51 lb
B. It is always 25 kg
C. There is no single universal OSHA maximum; assess the task and worker factors
D. It is 100 lb for men

Recall note: There is no single OSHA general-industry legal maximum weight that every worker may lift. The NIOSH RNLE uses a 51-lb load constant under ideal conditions and adjusts it with task multipliers to calculate RWL. The correct exam response depends on whether the question asks for the NIOSH load constant or a legal maximum.

44. Waste Reuse, Recycling and Recovery

Q87. Waste is sent offsite and processed into material that can be used again. This is BEST described as:

A. Recycling/reclamation
B. Direct reuse only
C. Abandonment
D. Dilution

Recall note: Reuse means using a material/product again without substantial reprocessing. Recycling generally processes/reclaims waste into usable material/product. Recovery extracts useful material or energy. Disposal is the final option. If offsite waste is processed/reclaimed into a usable material, recycling/reclamation is the likely answer.

Q88. A container is cleaned and used again for the same purpose without substantial reprocessing. This is:

A. Reuse
B. Incineration
C. Landfilling
D. Neutralization

Recall note: Reuse means using a material/product again without substantial reprocessing. Recycling generally processes/reclaims waste into usable material/product. Recovery extracts useful material or energy. Disposal is the final option. If offsite waste is processed/reclaimed into a usable material, recycling/reclamation is the likely answer.

45. Lockdown, Shelter-in-Place and Evacuation

Q89. An armed threat is nearby and movement through corridors could expose employees. Which protective action is MOST appropriate?

A. Lockout
B. Lockdown
C. Routine evacuation through the threat area
D. Hot-work permit

Recall note: Evacuation moves people away from a hazard. Shelter-in-place keeps people inside a safer location, commonly for external airborne/weather hazards. Lockdown restricts movement/access when movement could expose people to an immediate violent/security threat. Lockout is hazardous-energy control and is unrelated to protective-action decisions.

Q90. A toxic vapor cloud is outside the building and indoor protection is available. Which action may be appropriate?

A. Shelter-in-place
B. Lockout/tagout
C. Confined-space entry
D. Machine guarding

Recall note: Evacuation moves people away from a hazard. Shelter-in-place keeps people inside a safer location, commonly for external airborne/weather hazards. Lockdown restricts movement/access when movement could expose people to an immediate violent/security threat. Lockout is hazardous-energy control and is unrelated to protective-action decisions.

46. Chemical Spill Size-Up

Q91. During initial size-up of a chemical spill, which pair is MOST important to understand?

A. Volatility and reactivity
B. Employee vacation schedule and payroll
C. Paint color and supplier logo
D. Building age and parking count

Recall note: Initial spill assessment should identify the substance and hazards, quantity/concentration, physical state, location, exposure pathways, reactivity/incompatibilities, flammability, toxicity, volatility and environmental/weather conditions. SDS and emergency response information are key. Life safety is the first priority.

Q92. What is the first priority in a hazardous chemical spill response?

A. Life safety and responder protection
B. Production recovery
C. Media interviews
D. Cost accounting

Recall note: Initial spill assessment should identify the substance and hazards, quantity/concentration, physical state, location, exposure pathways, reactivity/incompatibilities, flammability, toxicity, volatility and environmental/weather conditions. SDS and emergency response information are key. Life safety is the first priority.

47. Sling Tension and Angle

Q93. For the same load, at which sling angle from the horizontal is leg tension greatest?

A. 90 degrees
B. 60 degrees
C. 45 degrees
D. 30 degrees

Recall note: For a symmetric two-leg sling with angle theta measured from the horizontal, leg tension T = W/(2 sin theta). As the sling angle decreases from horizontal, tension increases. At 90 degrees each leg carries W/2; at 60 degrees about 0.577W; at 45 degrees about 0.707W; at 30 degrees each leg is about W.

Q94. A 1,000-lb load is lifted by two equal sling legs at 30 degrees from horizontal. Approximate tension in EACH leg is:

A. 250 lb
B. 500 lb
C. 707 lb
D. 1,000 lb

Recall note: For a symmetric two-leg sling with angle theta measured from the horizontal, leg tension T = W/(2 sin theta). As the sling angle decreases from horizontal, tension increases. At 90 degrees each leg carries W/2; at 60 degrees about 0.577W; at 45 degrees about 0.707W; at 30 degrees each leg is about W.

48. TWA Exposure Calculation

Q95. A worker is exposed to 100 ppm for 4 hours and 0 ppm for the remaining 4 hours. What is the 8-hour TWA?

A. 25 ppm
B. 50 ppm
C. 75 ppm
D. 100 ppm

Recall note: TWA = sum(concentration x time) / total averaging time. For an 8-hour TWA, 4 hours at 100 ppm plus 4 hours at 0 ppm gives (100x4 + 0x4)/8 = 50 ppm. Always use consistent time units.

Q96. Which equation represents an 8-hour TWA for varying concentrations?

A. Sum(Ci x ti) / 8 h

B. Highest concentration only

C. Sum(Ci) x 8

D. Lowest concentration only

Recall note: TWA = sum(concentration x time) / total averaging time. For an 8-hour TWA, 4 hours at 100 ppm plus 4 hours at 0 ppm gives (100x4 + 0x4)/8 = 50 ppm. Always use consistent time units.

49. Halon and Ozone Depletion

Q97. Which legacy fire-extinguishing agent is strongly associated with stratospheric ozone depletion?

A. Water

B. Halon

C. AFFF water solution only

D. Class D salt only

Recall note: Halon fire-extinguishing agents are highly effective but are ozone-depleting substances; production/use has been heavily restricted and alternatives are preferred. Do not confuse ozone depletion with greenhouse warming alone.

Q98. The environmental concern that drove phaseout/restriction of many halons is primarily:

A. Ozone-layer depletion

B. Groundwater hardness

C. Noise pollution

D. Soil compaction

Recall note: Halon fire-extinguishing agents are highly effective but are ozone-depleting substances; production/use has been heavily restricted and alternatives are preferred. Do not confuse ozone depletion with greenhouse warming alone.

50. PPE Levels A-D

Q99. Which HAZWOPER PPE level provides the highest combined respiratory and skin protection?

A. Level A

B. Level B

C. Level C

D. Level D

Recall note: HAZWOPER PPE levels: Level A = highest respiratory plus highest skin protection (fully encapsulating suit with positive-pressure SCBA/SAR escape). Level B = highest respiratory protection but less skin protection. Level C = APR/PAPR when contaminants and concentrations are known and air-purifying criteria are met. Level D = basic work clothing/minimal protection.

Q100. Which level is selected when high respiratory protection is required but less skin protection is needed?

A. Level A

B. Level B

C. Level C

D. Level D

Recall note: HAZWOPER PPE levels: Level A = highest respiratory plus highest skin protection (fully encapsulating suit with positive-pressure SCBA/SAR escape). Level B = highest respiratory protection but less skin protection. Level C = APR/PAPR when contaminants and concentrations are known and air-purifying criteria are met. Level D = basic work clothing/minimal protection.

51. Benzene and Carcinogens

Q101. Which substance is a well-known occupational human carcinogen associated with leukemia?

A. Benzene

B. Nitrogen

C. Water vapor

D. Helium

Recall note: Benzene is a recognized human carcinogen strongly associated with hematologic effects, especially leukemia. In CSP questions, distinguish carcinogens (cancer), mutagens (genetic/DNA changes), teratogens (developmental effects), nephrotoxins (kidney) and hepatotoxins (liver).

Q102. A chemical that primarily causes heritable DNA changes is a:

A. Mutagen

B. Teratogen only

C. Nephrotoxin

D. Asphyxiant only

Recall note: Benzene is a recognized human carcinogen strongly associated with hematologic effects, especially leukemia. In CSP questions, distinguish carcinogens (cancer), mutagens (genetic/DNA changes), teratogens (developmental effects), nephrotoxins (kidney) and hepatotoxins (liver).

52. Asbestos Exposure Limits

Q103. What is OSHA's 8-hour TWA PEL for asbestos?

A. 0.01 f/cc

B. 0.1 f/cc

C. 1.0 f/cc

D. 10 f/cc

Recall note: OSHA general-industry asbestos PEL is 0.1 fiber per cubic centimeter (f/cc) as an 8-hour TWA. The excursion limit is 1.0 f/cc averaged over 30 minutes. Exposure monitoring uses representative breathing-zone samples.

Q104. What is OSHA's asbestos excursion limit?

A. 0.1 f/cc for 8 h

B. 1.0 f/cc averaged over 30 min

C. 5 f/cc for 15 min

D. 10 ppm ceiling

Recall note: OSHA general-industry asbestos PEL is 0.1 fiber per cubic centimeter (f/cc) as an 8-hour TWA. The excursion limit is 1.0 f/cc averaged over 30 minutes. Exposure monitoring uses representative breathing-zone samples.

53. Surface vs Grab Sampling

Q105. A CSP wants to determine contamination deposited on a work surface. Which method is MOST appropriate?

A. Surface/wipe sampling

B. Noise dosimetry

C. WBGT monitoring

D. Audiometry

Recall note: Surface/wipe sampling evaluates contamination deposited on a surface. Grab sampling is a short-duration sample taken at a particular time/location, often for air/gas or liquid characterization. If the question specifically says contamination on the ground surface, surface/wipe sampling is the stronger answer.

Q106. A one-time sample collected at a specific moment is generally called:

A. Grab sample

B. Full-shift integrated sample

C. Biological half-life

D. Area zoning

Recall note: Surface/wipe sampling evaluates contamination deposited on a surface. Grab sampling is a short-duration sample taken at a particular time/location, often for air/gas or liquid characterization. If the question specifically says contamination on the ground surface, surface/wipe sampling is the stronger answer.

54. ANSI/ASSP Z10 and PDCA

Q107. Which management cycle is closely associated with ANSI/ASSP Z10?

A. PDCA

B. 6M

C. RPN

D. LEL/UEL

Recall note: ANSI/ASSP Z10 is an occupational health and safety management-system standard built around continual improvement concepts consistent with Plan-Do-Check-Act. Auditing/monitoring performance is associated with Check; implementing planned controls is Do; corrective improvement is Act.

Q108. A CSP audits whether the safety-management system is operating as planned. Which PDCA phase is this?

A. Plan

B. Do

C. Check

D. Act

Recall note: ANSI/ASSP Z10 is an occupational health and safety management-system standard built around continual improvement concepts consistent with Plan-Do-Check-Act. Auditing/monitoring performance is associated with Check;

implementing planned controls is Do; corrective improvement is Act.

55. Workplace Violence Types

Q109. A stranger enters a workplace to commit robbery and threatens employees. Which workplace-violence type is this?

A. Type I - criminal intent

B. Type II - customer/client

C. Type III - worker-on-worker

D. Type IV - personal relationship

Recall note: Workplace violence is commonly grouped by relationship: Type I criminal intent (e.g., robbery by an outsider), Type II customer/client/patient, Type III worker-on-worker, Type IV personal relationship. Harassment, threats and stalking may fall into different types depending on the perpetrator relationship.

Q110. An employee threatens a coworker after a workplace dispute. This is MOST consistent with:

A. Type I

B. Type II

C. Type III

D. Type IV

Recall note: Workplace violence is commonly grouped by relationship: Type I criminal intent (e.g., robbery by an outsider), Type II customer/client/patient, Type III worker-on-worker, Type IV personal relationship. Harassment, threats and stalking may fall into different types depending on the perpetrator relationship.

56. Manual Handling and MSD Risk

Q111. What is the leading health risk associated with poorly designed manual material handling?

A. Musculoskeletal disorders

B. Ionizing radiation injury

C. Decompression sickness

D. Hearing loss only

Recall note: The main injury concern in manual material handling is musculoskeletal disorder (MSD), especially low-back and upper-extremity injury. Risk factors include high force, repetition, awkward posture, long duration, vibration and insufficient recovery.

Q112. Which combination MOST increases MSD risk?

A. Low force and neutral posture

B. High force, repetition and awkward posture

C. Adequate recovery and mechanical aids

D. Reduced reach distance

Recall note: The main injury concern in manual material handling is musculoskeletal disorder (MSD), especially low-back and upper-extremity injury. Risk factors include high force, repetition, awkward posture, long duration, vibration and insufficient recovery.

57. Kinesthetic Learning and U-Shaped Classroom

Q113. Which method is MOST suitable for a kinesthetic learner?

A. Hands-on practice

B. Silent reading only

C. Audio lecture only

D. Policy memo only

Recall note: Kinesthetic learners benefit from doing: demonstrations, hands-on practice, simulations and physical participation. A U-shaped classroom arrangement supports eye contact, discussion and trainer movement/access to participants.

Q114. Which classroom layout best supports trainer movement and direct interaction with participants?

A. U-shaped

B. Rows facing away from trainer

C. Closed cubicles

D. Random standing only

Recall note: Kinesthetic learners benefit from doing: demonstrations, hands-on practice, simulations and physical participation. A U-shaped classroom arrangement supports eye contact, discussion and trainer movement/access to participants.

58. Return on Investment (ROI)

Q115. A safety project costs $20,000 and produces $50,000 in measurable benefit. What is ROI using (Benefit-Cost)/Cost?

A. 60%

B. 100%

C. 150%

D. 250%

Recall note: Safety ROI compares financial benefits with investment cost. A common formula is ROI = (Benefit - Cost) / Cost x 100%. Include avoided losses/productivity gains when reasonably quantifiable, and distinguish tangible from intangible benefits.

Q116. The primary purpose of safety ROI analysis is to:

A. Compare economic benefit with investment cost

B. Replace legal compliance

C. Ignore intangible benefits

D. Calculate noise dose

Recall note: Safety ROI compares financial benefits with investment cost. A common formula is ROI = (Benefit - Cost) / Cost x 100%. Include avoided losses/productivity gains when reasonably quantifiable, and distinguish tangible from intangible benefits.

59. Temporary Power Tools: Grounding, Double Insulation and GFCI

Q117. A portable electric tool is not double-insulated. Which basic protection is important?

A. Proper equipment grounding

B. Removing the grounding pin

C. Wet hands

D. Bypassing GFCI

Recall note: Portable electric tools must be protected from electrical hazards. Depending on tool/design and environment, protection can include equipment grounding, approved double insulation and GFCI protection where required. A double-insulated tool does not use an equipment-grounding conductor in the same way as a grounded metal tool.

Q118. Which statement about double-insulated tools is correct?

A. They require removal of all guards

B. They provide an approved alternative design to equipment grounding for the tool

C. They can be used with damaged cords

D. They eliminate all electrical risk

Recall note: Portable electric tools must be protected from electrical hazards. Depending on tool/design and environment, protection can include equipment grounding, approved double insulation and GFCI protection where required. A double-insulated tool does not use an equipment-grounding conductor in the same way as a grounded metal tool.

60. Frequency: Hertz

Q119. A source operates at 60 Hz. How many cycles occur each second?

A. 1

B. 60

C. 600

D. 3,600

Recall note: Hertz (Hz) means cycles per second. Therefore 60 Hz = 60 cycles/second = 3,600 cycles/minute. Do not confuse frequency with rotational speed unless the system directly links them.

Q120. 60 cycles per second is equal to how many cycles per minute?

A. 60

B. 600

C. 3,600

D. 36,000

Recall note: Hertz (Hz) means cycles per second. Therefore 60 Hz = 60 cycles/second = 3,600 cycles/minute. Do not confuse frequency with rotational speed unless the system directly links them.

61. Abrasive / Cutting Wheel Speed Check

Q121. Before using an abrasive wheel, the machine spindle speed must:

A. Exceed the wheel's rated maximum speed

B. Not exceed the wheel's rated maximum speed

C. Be ignored

D. Always be half the wheel rating

Recall note: Before mounting/using an abrasive wheel, verify the wheel's rated maximum RPM is at least the machine spindle speed, inspect/ring-test where applicable, use the proper guard and stand clear during startup. OSHA requires certain bench/floor stand wheels to be run at operating speed with guards in place for at least one minute before applying work. '20 seconds' is not the general OSHA rule.

Q122. For a newly mounted wheel on applicable bench/floor stand grinders, OSHA requires a test run at operating speed for at least:

A. 10 seconds

B. 20 seconds

C. 1 minute

D. 10 minutes

Recall note: Before mounting/using an abrasive wheel, verify the wheel's rated maximum RPM is at least the machine spindle speed, inspect/ring-test where applicable, use the proper guard and stand clear during startup. OSHA requires certain bench/floor stand wheels to be run at operating speed with guards in place for at least one minute before applying work. '20 seconds' is not the general OSHA rule.

62. Fire Load Calculation

Q123. Which formula BEST represents fire-load density?

A. Sum(mass x heat of combustion) / floor area

B. Floor area / combustible mass

C. Temperature x time

D. Mass / sprinkler count

Recall note: Fire load is the potential heat energy available from combustible contents. Fire-load density = sum(m_i x H_i) / floor area, where m is combustible mass and H is heat of combustion. Units may be MJ/m2 or equivalent.

Q124. A room contains more combustible material but the same floor area. What happens to fire-load density?

A. It generally increases

B. It must decrease

C. It becomes zero

D. It is unrelated to combustibles

Recall note: Fire load is the potential heat energy available from combustible contents. Fire-load density = sum(m_i x H_i) / floor area, where m is combustible mass and H is heat of combustion. Units may be MJ/m2 or equivalent.

63. Resistance of a Conductor

Q125. For the same material and cross-sectional area, increasing conductor length will:

A. Increase resistance

B. Decrease resistance

C. Not affect resistance

D. Make resistance zero

Recall note: Electrical resistance R = rho L/A. For the same material and temperature, resistance increases with conductor length and decreases with cross-sectional area. Ohm's law is V = IR.

Q126. Which relationship is Ohm's law?

A. V = IR

B. Q = VA

C. F = ma

D. P = F(1+i)^n

Recall note: Electrical resistance R = rho L/A. For the same material and temperature, resistance increases with conductor length and decreases with cross-sectional area. Ohm's law is V = IR.

64. Mutagen, Teratogen, Nephrotoxin, Hepatotoxin

Q127. A laboratory chemical causes mutations in genetic material. It is classified as a:

A. Mutagen

B. Teratogen

C. Nephrotoxin

D. Simple asphyxiant

Recall note: Mutagen = genetic/DNA mutation; teratogen = adverse embryo/fetal development; nephrotoxin = kidney damage; hepatotoxin = liver damage. Carcinogen causes or increases cancer risk. The recalled 'laboratory worker feels changes in his body' wording is too vague by itself; choose mutagen only when genetic mutation is the clue.

Q128. A substance that primarily damages the kidneys is a:

A. Hepatotoxin
B. Nephrotoxin
C. Teratogen
D. Sensitizer only

Recall note: Mutagen = genetic/DNA mutation; teratogen = adverse embryo/fetal development; nephrotoxin = kidney damage; hepatotoxin = liver damage. Carcinogen causes or increases cancer risk. The recalled 'laboratory worker feels changes in his body' wording is too vague by itself; choose mutagen only when genetic mutation is the clue.

65. Qualitative vs Quantitative Risk Analysis

Q129. A risk matrix using Low, Medium and High categories is primarily:

A. Qualitative
B. Quantitative only
C. Financial accounting
D. Epidemiology

Recall note: Qualitative risk analysis uses descriptive categories and expert judgment (e.g., low/medium/high). Quantitative analysis uses numerical probabilities, frequencies, consequence values or modeled loss estimates. Semi-quantitative methods assign scores to categories but are not fully probabilistic.

Q130. A study estimates an annual fatality probability of 2 x 10^-5. This is:

A. Qualitative
B. Quantitative
C. Role play
D. Brainstorming

Recall note: Qualitative risk analysis uses descriptive categories and expert judgment (e.g., low/medium/high). Quantitative analysis uses numerical probabilities, frequencies, consequence values or modeled loss estimates. Semi-quantitative methods assign scores to categories but are not fully probabilistic.

66. Safety Equipment in ICS

Q131. Who is MOST directly responsible for monitoring responder safety conditions in ICS?

A. Safety Officer
B. Liaison Officer
C. Finance Section Chief
D. PIO

Recall note: Within ICS, the Safety Officer monitors safety conditions and develops safety measures for responders. The Incident Commander remains overall responsible for the incident. When the exam asks who ensures responder safety measures/equipment are addressed, Safety Officer is usually the targeted role unless wording explicitly asks for overall command responsibility.

Q132. Who retains overall responsibility for the incident even when a Safety Officer is appointed?

A. Incident Commander
B. Liaison Officer
C. Staging Manager
D. Technical specialist

Recall note: Within ICS, the Safety Officer monitors safety conditions and develops safety measures for responders. The Incident Commander remains overall responsible for the incident. When the exam asks who ensures responder safety measures/equipment are addressed, Safety Officer is usually the targeted role unless wording explicitly asks for overall command responsibility.

67. Pareto 80/20 Principle

Q133. Which tool is associated with the 'vital few' and the 80/20 principle?

A. Pareto analysis
B. HAZOP
C. FMEA
D. JSA

Recall note: Pareto analysis ranks problems/causes by frequency or impact so attention can focus on the vital few. The 80/20 relationship is a rule of thumb, not a physical law: roughly 80% of effects may come from 20% of causes.

Q134. The BEST use of a Pareto chart in safety is to:

A. Prioritize the categories contributing most to the problem

B. Calculate sling tension

C. Measure noise frequency

D. Select respirator cartridges

Recall note: Pareto analysis ranks problems/causes by frequency or impact so attention can focus on the vital few. The 80/20 relationship is a rule of thumb, not a physical law: roughly 80% of effects may come from 20% of causes.

68. Safety Through Design and Retrofitting

Q135. A key economic advantage of designing safety into a new system is:

A. Avoiding costly later retrofits

B. Increasing reliance on PPE

C. Delaying hazard identification

D. Eliminating maintenance planning

Recall note: Design-stage hazard control is usually more effective and economical than adding controls after installation. PtD seeks to eliminate hazards or design in controls before equipment/facilities are built, thereby reducing later retrofit costs and constraints.

Q136. Which control should generally be considered earliest during design?

A. Eliminate the hazard where feasible

B. Write a warning label only

C. Issue PPE only

D. Accept every risk

Recall note: Design-stage hazard control is usually more effective and economical than adding controls after installation. PtD seeks to eliminate hazards or design in controls before equipment/facilities are built, thereby reducing later retrofit costs and constraints.

69. Fire Prevention and Housekeeping

Q137. Why is housekeeping important to fire prevention?

A. It reduces combustible accumulation and keeps access/egress clear

B. It increases fire load

C. It replaces sprinklers

D. It eliminates the need for alarms

Recall note: Good housekeeping removes unnecessary combustible loading, keeps exits/access clear and reduces ignition/fuel opportunities. It is a proactive/leading activity when measured as inspections or corrective-action completion.

Q138. Percentage of scheduled housekeeping inspections completed is a:

A. Leading indicator

B. Lagging indicator

C. Case-fatality rate

D. APF

Recall note: Good housekeeping removes unnecessary combustible loading, keeps exits/access clear and reduces ignition/fuel opportunities. It is a proactive/leading activity when measured as inspections or corrective-action completion.

70. GHS Label Elements

Q139. Which item is required on a GHS/HazCom shipped-container label?

A. Product identifier

B. Atomic number

C. Employee ID

D. Purchase price

Recall note: OSHA HazCom shipped-container labels include product identifier, signal word, hazard statement(s), pictogram(s), precautionary statement(s), and responsible-party name/address/telephone. Signal word and hazard statement are not interchangeable.

Q140. Which label element tells the user recommended measures to minimize/prevent adverse effects?

A. Precautionary statement

B. Signal word

C. Product identifier

D. CAS number only

Recall note: OSHA HazCom shipped-container labels include product identifier, signal word, hazard statement(s), pictogram(s), precautionary statement(s), and responsible-party name/address/telephone. Signal word and hazard statement are not interchangeable.

71. Industrial Hygiene: Anticipate, Recognize, Evaluate, Control

Q141. Which ventilation method captures contaminants at or near the source?

A. Local exhaust ventilation

B. General dilution only

C. Natural lighting

D. Radiant heating

Recall note: Industrial hygiene protects worker health by anticipating and recognizing hazards, evaluating exposures, and controlling them. Some study materials add confirmation/verification after control. LEV captures contaminants near the source; dilution ventilation reduces concentration by mixing with cleaner air.

Q142. What normally follows recognition of an occupational health hazard in the IH process?

A. Evaluation of exposure

B. Ignoring it

C. Payroll review

D. Fire-load calculation

Recall note: Industrial hygiene protects worker health by anticipating and recognizing hazards, evaluating exposures, and controlling them. Some study materials add confirmation/verification after control. LEV captures contaminants near the source; dilution ventilation reduces concentration by mixing with cleaner air.

72. Confined Space / IDLH

Q143. Which characteristic is part of OSHA's confined-space definition?

A. Designed for continuous occupancy

B. Limited or restricted means of entry or exit

C. Always underground

D. Always contains toxic gas

Recall note: A confined space is large enough to enter, has limited/restricted entry or exit, and is not designed for continuous occupancy. A permit-required confined space has additional serious hazards such as hazardous atmosphere, engulfment, entrapment configuration or other recognized serious hazard. IDLH means immediately dangerous to life or health.

Q144. IDLH means:

A. Immediately Dangerous to Life or Health

B. Industrial Design Load Hazard

C. Internal Daily Limit for Health

D. Incident Data Loss History

Recall note: A confined space is large enough to enter, has limited/restricted entry or exit, and is not designed for continuous occupancy. A permit-required confined space has additional serious hazards such as hazardous atmosphere, engulfment, entrapment configuration or other recognized serious hazard. IDLH means immediately dangerous to life or health.

73. TLV-TWA, STEL and Ceiling

Q145. Which TLV is a 15-minute short-term exposure limit?

A. TLV-TWA

B. TLV-STEL

C. TLV-C only

D. IDLH

Recall note: ACGIH TLVs are guidelines, not OSHA PELs. TLV-TWA is an 8-hour/day, 40-hour/week time-weighted average framework; TLV-STEL is a 15-minute short-term exposure limit; TLV-C is a ceiling that should not be exceeded at any time.

Q146. Which limit should not be exceeded at any time?

A. TLV-C

B. 8-hour TWA only

C. Action level only

D. RWL

Recall note: ACGIH TLVs are guidelines, not OSHA PELs. TLV-TWA is an 8-hour/day, 40-hour/week time-weighted average framework; TLV-STEL is a 15-minute short-term exposure limit; TLV-C is a ceiling that should not be exceeded at any time.

74. LD50 and LC50

Q147. LD50 refers to:

A. Lethal dose for 50%
B. Legal dose for 50 hours
C. Lowest detectable concentration
D. Lifting demand 50

Recall note: LD50 is the dose expected to cause death in 50% of a test population under specified conditions; LC50 is the concentration expected to cause death in 50%. Lower LD50/LC50 generally indicates greater acute toxicity when comparing comparable routes/species/conditions.

Q148. For otherwise comparable substances, a lower LD50 generally indicates:

A. Greater acute toxicity
B. Lower acute toxicity
C. No toxicity
D. Higher flash point

Recall note: LD50 is the dose expected to cause death in 50% of a test population under specified conditions; LC50 is the concentration expected to cause death in 50%. Lower LD50/LC50 generally indicates greater acute toxicity when comparing comparable routes/species/conditions.

75. Behavior-Based Safety (BBS)

Q149. Which activity is MOST characteristic of a BBS program?

A. Observing defined behaviors and providing feedback
B. Replacing machine guards with warnings
C. Ignoring system causes
D. Calculating fire load

Recall note: BBS focuses on observable behavior, antecedents and consequences, feedback and reinforcement. It can support proactive safety improvement but should not be used to blame workers or replace correction of system/design hazards.

Q150. A sound BBS program should:

A. Complement system and hazard controls
B. Blame workers for all incidents
C. Replace engineering controls
D. Eliminate incident investigation

Recall note: BBS focuses on observable behavior, antecedents and consequences, feedback and reinforcement. It can support proactive safety improvement but should not be used to blame workers or replace correction of system/design hazards.

76. Top Management Commitment

Q151. Who is ultimately responsible for establishing organizational direction and commitment for the safety-management system?

A. Top management
B. Safety technician alone
C. Temporary worker
D. External auditor

Recall note: Effective safety management requires top management to establish policy/direction, provide resources, assign roles/accountability, integrate safety into business processes, review performance and demonstrate visible commitment. Safety is not solely the safety department's responsibility.

Q152. Which action BEST demonstrates top-management commitment?

A. Providing resources and integrating safety into business decisions
B. Delegating all safety responsibility to HSE and disengaging
C. Reviewing only injury statistics once a year
D. Ignoring corrective actions

Recall note: Effective safety management requires top management to establish policy/direction, provide resources, assign roles/accountability, integrate safety into business processes, review performance and demonstrate visible commitment. Safety is not solely the safety department's responsibility.

77. Risk Transfer and Retention

Q153. Purchasing insurance is an example of:

A. Risk transfer
B. Risk avoidance
C. Risk elimination
D. Risk retention only

Recall note: Risk transfer shifts financial consequences to another party, commonly through insurance or contractual allocation/indemnity, though responsibility may not disappear. Risk retention means knowingly keeping/accepting the risk and financing consequences if loss occurs.

Q154. A company knowingly accepts a low-level risk and budgets for possible loss. This is:

A. Risk retention
B. Risk transfer
C. Subrogation
D. Elimination

Recall note: Risk transfer shifts financial consequences to another party, commonly through insurance or contractual allocation/indemnity, though responsibility may not disappear. Risk retention means knowingly keeping/accepting the risk and financing consequences if loss occurs.

78. Strict, Contributory and Comparative Liability Concepts

Q155. A doctrine that may impose liability without proving negligence is:

A. Strict liability
B. Delegation
C. Subrogation
D. Risk retention

Recall note: Strict liability can impose liability without proving negligence in specified activities/products. Contributory negligence can bar a plaintiff's recovery in jurisdictions using that doctrine if the plaintiff contributed to the harm. Comparative negligence apportions damages according to fault. These are legal concepts and jurisdiction-specific.

Q156. A system that apportions damages according to each party's percentage of fault is:

A. Comparative negligence
B. Strict liability
C. Workers' compensation only
D. Respondeat superior only

Recall note: Strict liability can impose liability without proving negligence in specified activities/products. Contributory negligence can bar a plaintiff's recovery in jurisdictions using that doctrine if the plaintiff contributed to the harm. Comparative negligence apportions damages according to fault. These are legal concepts and jurisdiction-specific.

79. Fire Sprinkler Systems

Q157. Which sprinkler system is commonly selected for areas subject to freezing?

A. Wet pipe
B. Dry pipe
C. Domestic hose only
D. Standpipe only

Recall note: Wet pipe contains pressurized water and is common where freezing is not a concern. Dry pipe contains pressurized air/nitrogen and admits water after activation, making it suitable for freezing areas. Deluge uses open sprinklers/nozzles and a detection-operated valve so many outlets discharge together. Preaction requires detection before water enters piping and is used where accidental discharge is especially undesirable.

Q158. Which system typically has open discharge outlets and releases water through all outlets when the deluge valve opens?

A. Wet pipe
B. Dry pipe
C. Deluge
D. Preaction only after individual bulb activation

Recall note: Wet pipe contains pressurized water and is common where freezing is not a concern. Dry pipe contains pressurized air/nitrogen and admits water after activation, making it suitable for freezing areas. Deluge uses open sprinklers/nozzles and a detection-operated valve so many outlets discharge together. Preaction requires detection before water enters piping and is used where accidental discharge is especially undesirable.

80. Emergency / Fire Alarm vs Ventilation vs Illumination

Q159. Which system's primary purpose is to warn employees of a fire or other emergency?

A. Employee alarm system
B. General ventilation
C. Normal task lighting
D. Forklift horn only

Recall note: Emergency systems serve different functions. Alarm warns occupants; emergency lighting supports safe egress when normal lighting fails; ventilation controls airborne contaminants/smoke depending on system design. If a question asks what warns occupants of an emergency, choose alarm; if it asks what permits visibility during power loss, choose emergency lighting.

Q160. Which safeguard primarily supports visibility along an exit route during loss of normal lighting?

A. Emergency lighting
B. LEV
C. SCBA
D. Grounding conductor

Recall note: Emergency systems serve different functions. Alarm warns occupants; emergency lighting supports safe egress when normal lighting fails; ventilation controls airborne contaminants/smoke depending on system design. If a question asks what warns occupants of an emergency, choose alarm; if it asks what permits visibility during power loss, choose emergency lighting.

81. Business Continuity vs Emergency Response

Q161. Which plan focuses MOST on maintaining critical business functions during disruption?

A. Business continuity plan
B. Hot-work permit
C. JSA
D. Respirator fit test

Recall note: Emergency response protects life, property and environment during the immediate event. Business continuity focuses on sustaining/restoring critical operations. Disaster recovery often emphasizes restoration of systems/IT/infrastructure after disruption. Crisis management provides strategic coordination and communication.

Q162. Immediate life-safety actions during a fire are primarily part of:

A. Emergency response
B. Long-term business continuity only
C. ROI analysis
D. Procurement

Recall note: Emergency response protects life, property and environment during the immediate event. Business continuity focuses on sustaining/restoring critical operations. Disaster recovery often emphasizes restoration of systems/IT/infrastructure after disruption. Crisis management provides strategic coordination and communication.

82. Training Needs Assessment and Evaluation

Q163. What is the fundamental purpose of a training needs assessment?

A. Determine performance gaps and whether training is an appropriate solution
B. Select lunch menus
C. Calculate fire load
D. Replace MOC

Recall note: Training needs assessment identifies performance gaps and determines whether training is an appropriate solution. Training cannot fix problems caused solely by poor equipment/design, inadequate resources or conflicting systems. Effectiveness should be evaluated against learning/performance objectives; practical skills are best checked by demonstration.

Q164. After fall-protection training, what BEST verifies a worker can correctly use the equipment?

A. Practical demonstration
B. Attendance sheet only
C. Course title
D. Training-room size

Recall note: Training needs assessment identifies performance gaps and determines whether training is an appropriate solution. Training cannot fix problems caused solely by poor equipment/design, inadequate resources or conflicting systems. Effectiveness should be evaluated against learning/performance objectives; practical skills are best checked by demonstration.

83. SWOT Analysis

Q165. What does SWOT stand for?

A. Strengths, Weaknesses, Opportunities, Threats
B. Safety, Work, Operations, Training
C. Systems, Waste, Objectives, Tactics
D. Standards, Work, Oversight, Testing

Recall note: SWOT = Strengths, Weaknesses, Opportunities, Threats. Strengths/weaknesses are usually internal; opportunities/threats are usually external. It supports strategic planning and situation assessment.

Q166. Which SWOT element is generally external and favorable?

A. Strength
B. Weakness
C. Opportunity
D. Internal control

Recall note: SWOT = Strengths, Weaknesses, Opportunities, Threats. Strengths/weaknesses are usually internal; opportunities/threats are usually external. It supports strategic planning and situation assessment.

84. TRIR and DART

Q167. Which constant is used in OSHA TRIR and DART rate calculations?

A. 1,000
B. 10,000
C. 100,000
D. 200,000

Recall note: OSHA incident rates normalize cases to 200,000 work hours (100 full-time workers x 40 h/week x 50 weeks). TRIR = recordable cases x 200,000 / hours worked. DART uses cases involving days away, restricted work or job transfer.

Q168. Which cases are used in the DART numerator?

A. Days away, restricted work or job transfer cases
B. First-aid cases only
C. Near misses only
D. All property damage

Recall note: OSHA incident rates normalize cases to 200,000 work hours (100 full-time workers x 40 h/week x 50 weeks). TRIR = recordable cases x 200,000 / hours worked. DART uses cases involving days away, restricted work or job transfer.

85. Pyrophoric and Water-Reactive Hazards

Q169. A chemical that can ignite spontaneously on contact with air is:

A. Pyrophoric
B. Inert
C. Cryogenic only
D. Noncombustible

Recall note: Pyrophoric materials can ignite spontaneously in air under specified conditions. Water-reactive materials can react violently with water, generating heat, flammable/toxic gases or other hazards. A material can have more than one hazard classification.

Q170. A water-reactive material may:

A. Generate heat or hazardous/flammable gas on contact with water
B. Always be safely extinguished with water
C. Have no chemical reaction
D. Only create noise

Recall note: Pyrophoric materials can ignite spontaneously in air under specified conditions. Water-reactive materials can react violently with water, generating heat, flammable/toxic gases or other hazards. A material can have more than one hazard classification.

86. WBGT and Heat Stress

Q171. Which instrument/component reflects radiant heat in WBGT assessment?

A. Globe temperature
B. Dry bulb only
C. Sound level meter

D. Anemometer only

Recall note: WBGT estimates environmental heat stress. Outdoor/direct-sun WBGT uses natural wet bulb, globe and dry bulb components; indoor/no-solar-load formula emphasizes natural wet bulb and globe. Wet bulb reflects evaporative/humidity effects; globe reflects radiant heat; dry bulb reflects air temperature. Workload and clothing also affect heat strain.

Q172. Occupational heat stress is influenced by:

A. Environmental heat, metabolic workload and clothing

B. Temperature only

C. Humidity only

D. Age only

Recall note: WBGT estimates environmental heat stress. Outdoor/direct-sun WBGT uses natural wet bulb, globe and dry bulb components; indoor/no-solar-load formula emphasizes natural wet bulb and globe. Wet bulb reflects evaporative/humidity effects; globe reflects radiant heat; dry bulb reflects air temperature. Workload and clothing also affect heat strain.

87. Fall Protection: Passive Before Active

Q173. Which is a passive fall-protection method?

A. Guardrail system

B. Personal fall-arrest harness

C. Self-retracting lifeline

D. Shock-absorbing lanyard

Recall note: Where feasible, eliminate/prevent the fall hazard before relying on personal fall arrest. Guardrails and covers are passive systems because they do not require worker action after installation; harness/lanyard systems are active. Fall arrest limits consequences after a fall begins; fall restraint/prevention prevents reaching the edge.

Q174. From a hierarchy/PtD perspective, which is generally preferred?

A. Prevent the fall with passive/design controls where feasible

B. Rely only on fall arrest

C. Remove guardrails

D. Increase free-fall distance

Recall note: Where feasible, eliminate/prevent the fall hazard before relying on personal fall arrest. Guardrails and covers are passive systems because they do not require worker action after installation; harness/lanyard systems are active. Fall arrest limits consequences after a fall begins; fall restraint/prevention prevents reaching the edge.

88. SDS Sections

Q175. Which SDS section covers exposure controls and personal protection?

A. Section 4

B. Section 6

C. Section 8

D. Section 11

Recall note: OSHA HazCom SDS uses 16 standardized sections. Key exam sections: 2 hazard identification; 4 first aid; 5 firefighting; 6 accidental release; 7 handling/storage; 8 exposure controls/PPE; 9 physical/chemical properties; 10 stability/reactivity; 11 toxicology. Sections 12-15 are included for GHS consistency but OSHA does not enforce their content.

Q176. Which SDS section covers accidental release measures?

A. Section 2

B. Section 6

C. Section 10

D. Section 16

Recall note: OSHA HazCom SDS uses 16 standardized sections. Key exam sections: 2 hazard identification; 4 first aid; 5 firefighting; 6 accidental release; 7 handling/storage; 8 exposure controls/PPE; 9 physical/chemical properties; 10 stability/reactivity; 11 toxicology. Sections 12-15 are included for GHS consistency but OSHA does not enforce their content.

89. Lockout/Tagout and Zero Energy

Q177. After locks/tags are applied and stored energy is controlled, what should occur before servicing?

A. Verify isolation/zero-energy condition

B. Immediately begin work without checking

C. Remove all locks

D. Energize for convenience

Recall note: LOTO controls hazardous energy during servicing/maintenance. Core sequence includes prepare, shut down, isolate, apply lock/tag, control stored energy, verify isolation, perform work, and restore safely. Verification of zero/isolated energy before work is critical.

Q178. LOTO primarily protects workers from:

A. Unexpected energization/startup or release of stored energy

B. Routine noise only

C. Sun exposure

D. Traffic congestion

Recall note: LOTO controls hazardous energy during servicing/maintenance. Core sequence includes prepare, shut down, isolate, apply lock/tag, control stored energy, verify isolation, perform work, and restore safely. Verification of zero/isolated energy before work is critical.

90. Ames Test

Q179. The Ames test is primarily used to screen for:

A. Mutagenicity

B. Noise-induced hearing loss

C. Heat stress

D. Ergonomic lifting risk

Recall note: The Ames test uses specially selected bacteria to screen chemicals for mutagenic activity. A positive result suggests mutation potential and possible genotoxic/carcinogenic concern, but it is a screening test, not definitive proof that a chemical causes cancer in humans.

Q180. The Ames test commonly uses:

A. Bacteria

B. Forklifts

C. Sprinklers

D. Sound-level meters

Recall note: The Ames test uses specially selected bacteria to screen chemicals for mutagenic activity. A positive result suggests mutation potential and possible genotoxic/carcinogenic concern, but it is a screening test, not definitive proof that a chemical causes cancer in humans.

91. Carpal Tunnel Syndrome

Q181. Which nerve is compressed in carpal tunnel syndrome?

A. Median nerve

B. Ulnar nerve

C. Sciatic nerve

D. Optic nerve

Recall note: Carpal tunnel syndrome results from compression of the median nerve at the wrist. Symptoms include numbness, tingling, pain and weakness, often affecting the thumb, index, middle and part of the ring finger. Repetition, force and awkward wrist posture can contribute.

Q182. Which ergonomic factors can contribute to CTS risk?

A. Repetition, force and awkward wrist posture

B. Only high temperature

C. Only radiation

D. Only noise

Recall note: Carpal tunnel syndrome results from compression of the median nerve at the wrist. Symptoms include numbness, tingling, pain and weakness, often affecting the thumb, index, middle and part of the ring finger. Repetition, force and awkward wrist posture can contribute.

92. Acid-Base Neutralization

Q183. A typical acid-base neutralization reaction produces:

A. Salt and water

B. Only oxygen

C. Only nitrogen

D. Metal dust

Recall note: An acid reacting with a base generally produces a salt and water in a neutralization reaction, though actual products depend on the chemicals and conditions. pH below 7 is acidic and above 7 is basic at ordinary aqueous conditions.

Q184. A solution with pH 3 is:

A. Acidic
B. Neutral
C. Basic
D. Always nonhazardous

Recall note: An acid reacting with a base generally produces a salt and water in a neutralization reaction, though actual products depend on the chemicals and conditions. pH below 7 is acidic and above 7 is basic at ordinary aqueous conditions.

93. Grounding vs Bonding

Q185. What is the primary purpose of bonding two containers during flammable-liquid transfer?

A. Equalize electrical potential between them
B. Increase liquid pressure
C. Cool the liquid
D. Measure pH

Recall note: Bonding electrically connects conductive objects to equalize potential and reduce sparking between them. Grounding connects an object/system to earth or a grounding system. During flammable-liquid transfer, bonding between containers is critical; grounding may also be required depending on the operation.

Q186. Grounding primarily provides:

A. A conductive connection to earth/grounding system
B. A chemical neutralizer
C. A fire-class label
D. A lifting point

Recall note: Bonding electrically connects conductive objects to equalize potential and reduce sparking between them. Grounding connects an object/system to earth or a grounding system. During flammable-liquid transfer, bonding between containers is critical; grounding may also be required depending on the operation.

94. Dust Explosion Pentagon

Q187. Which two elements are added to the fire triangle to form the dust-explosion pentagon?

A. Dispersion and confinement
B. Pressure and temperature
C. Water and foam
D. Grounding and bonding

Recall note: A dust explosion requires the fire-triangle elements - fuel, ignition source and oxygen - plus dispersion of combustible dust and confinement. Removing any required element can prevent the explosion. Good housekeeping helps control fuel accumulation.

Q188. Which control directly reduces the combustible-dust fuel available?

A. Effective housekeeping/dust collection
B. Increasing dust dispersion
C. Adding confinement
D. Creating ignition sources

Recall note: A dust explosion requires the fire-triangle elements - fuel, ignition source and oxygen - plus dispersion of combustible dust and confinement. Removing any required element can prevent the explosion. Good housekeeping helps control fuel accumulation.

95. Flash Point and LEL/UEL

Q189. Flash point is MOST closely related to:

A. The temperature at which sufficient vapor is produced for ignition under test conditions
B. The temperature at which all liquids boil
C. The freezing point
D. The autoignition temperature only

Recall note: Flash point is the lowest temperature at which a liquid gives off enough vapor to form an ignitable mixture near its surface under specified test conditions. Below the LEL a mixture is too lean; above the UEL it is too rich; between LEL and UEL it is within the flammable range.

Q190. A vapor concentration below its LEL is:

A. Too lean to burn
B. Too rich to burn

C. Always explosive

D. At stoichiometric concentration

Recall note: Flash point is the lowest temperature at which a liquid gives off enough vapor to form an ignitable mixture near its surface under specified test conditions. Below the LEL a mixture is too lean; above the UEL it is too rich; between LEL and UEL it is within the flammable range.

96. Hierarchy of Controls

Q191. Which control is highest in the hierarchy?

A. Elimination

B. Engineering control

C. Administrative control

D. PPE

Recall note: The standard hierarchy is elimination, substitution, engineering controls, administrative/work-practice controls, then PPE. Engineering controls isolate people from hazards; PPE is the last line of defense. Do not start with engineering if elimination or substitution is feasible.

Q192. Installing local exhaust ventilation to capture solvent vapor is:

A. Engineering control

B. Administrative control

C. PPE

D. Risk transfer

Recall note: The standard hierarchy is elimination, substitution, engineering controls, administrative/work-practice controls, then PPE. Engineering controls isolate people from hazards; PPE is the last line of defense. Do not start with engineering if elimination or substitution is feasible.

97. Emergency Bomb Threat - Initial Actions

Q193. An employee receives a bomb-threat call. What is the BEST immediate approach?

A. Follow the bomb-threat procedure, preserve caller information and promptly notify designated authorities

B. Hang up and ignore it

C. Personally search and handle suspicious packages

D. Broadcast unverified details on social media

Recall note: Bomb-threat response follows the organization's emergency/security procedure. Preserve information from the caller/message, notify designated security/emergency authorities promptly, avoid actions that could create additional risk, and do not independently search/handle suspicious objects unless trained/assigned. The exact 'first' action depends on how the threat is received and site procedure.

Q194. Who should normally decide protective actions such as evacuation based on the threat assessment?

A. Designated incident/security authorities under the emergency plan

B. Any individual employee acting alone

C. The payroll clerk only

D. Visitors

Recall note: Bomb-threat response follows the organization's emergency/security procedure. Preserve information from the caller/message, notify designated security/emergency authorities promptly, avoid actions that could create additional risk, and do not independently search/handle suspicious objects unless trained/assigned. The exact 'first' action depends on how the threat is received and site procedure.

98. Emergency Exercises

Q195. A team discusses a simulated disaster scenario around a table without deploying field resources. This is a:

A. Tabletop exercise

B. Full-scale exercise

C. Live-fire drill

D. Medical surveillance exam

Recall note: Discussion-based exercises include seminars, workshops, tabletop exercises and games. Operations-based exercises include drills, functional exercises and full-scale exercises. A tabletop is useful for discussing roles, decisions and coordination without deploying full field resources.

Q196. Which is operations-based rather than discussion-based?

A. Drill

B. Seminar

C. Workshop

D. Tabletop

Recall note: Discussion-based exercises include seminars, workshops, tabletop exercises and games. Operations-based exercises include drills, functional exercises and full-scale exercises. A tabletop is useful for discussing roles, decisions and coordination without deploying full field resources.

99. Biosafety Levels

Q197. Which biosafety level is appropriate for agents with minimal potential hazard to healthy adults?

A. BSL-1

B. BSL-2

C. BSL-3

D. BSL-4

Recall note: BSL-1 is for well-characterized agents not known to consistently cause disease in healthy adults and represents minimal potential hazard. BSL-2 is moderate hazard; BSL-3 is for agents that may cause serious/potentially lethal disease through inhalation; BSL-4 is for dangerous/exotic agents with high individual risk and often no available treatment/vaccine.

Q198. Which level is associated with serious or potentially lethal disease through inhalation and enhanced containment?

A. BSL-1

B. BSL-2

C. BSL-3

D. Office level

Recall note: BSL-1 is for well-characterized agents not known to consistently cause disease in healthy adults and represents minimal potential hazard. BSL-2 is moderate hazard; BSL-3 is for agents that may cause serious/potentially lethal disease through inhalation; BSL-4 is for dangerous/exotic agents with high individual risk and often no available treatment/vaccine.

100. Q = V x A Airflow

Q199. Air velocity is 500 ft/min through a 2 ft2 duct. What is airflow?

A. 250 cfm

B. 500 cfm

C. 1,000 cfm

D. 2,000 cfm

Recall note: Volumetric flow Q equals average air velocity V times cross-sectional area A. In U.S. ventilation work, if V is ft/min and A is ft2, Q is ft3/min (cfm). For a circular duct, A = pi D2/4.

Q200. Which equation gives volumetric airflow?

A. Q = V x A

B. V = IR

C. F = kx

D. E = mc2

Recall note: Volumetric flow Q equals average air velocity V times cross-sectional area A. In U.S. ventilation work, if V is ft/min and A is ft2, Q is ft3/min (cfm). For a circular duct, A = pi D2/4.

101. Hooke's Law

Q201. A spring constant is 200 N/m and displacement is 0.05 m. Force is:

A. 4 N

B. 10 N

C. 40 N

D. 4,000 N

Recall note: Hooke's law for an ideal linear spring is F = kx, where F is force, k is spring constant and x is displacement from equilibrium. It applies within the elastic/linear range.

Q202. In F = kx, k represents:

A. Spring constant

B. Kinetic energy

C. Temperature

D. Resistance

Recall note: Hooke's law for an ideal linear spring is F = kx, where F is force, k is spring constant and x is displacement from equilibrium. It applies within the elastic/linear range.

102. Mean, Median, Mode and Standard Deviation

Q203. For a normal distribution with mean 68 and SD 15, about 95% of values lie approximately between:

A. 53 and 83

B. 38 and 98

C. 23 and 113

D. 68 and 83

Recall note: Mean is arithmetic average; median is the middle ordered value; mode is most frequent; range is max-min. Standard deviation describes spread around the mean. For an approximately normal distribution: about 68% lies within +/-1 SD, 95% within +/-2 SD and 99.7% within +/-3 SD.

Q204. Which statistic describes the spread of values around the mean?

A. Standard deviation

B. Mode only

C. Median only

D. Sample size only

Recall note: Mean is arithmetic average; median is the middle ordered value; mode is most frequent; range is max-min. Standard deviation describes spread around the mean. For an approximately normal distribution: about 68% lies within +/-1 SD, 95% within +/-2 SD and 99.7% within +/-3 SD.

103. ALARA and Radiation Protection

Q205. Which three controls are fundamental to external radiation protection?

A. Time, distance, shielding

B. Heat, humidity, wind

C. Elimination, payroll, insurance

D. Voltage, current, resistance

Recall note: ALARA means As Low As Reasonably Achievable. Radiation dose is reduced by minimizing time, maximizing distance and using shielding. For a point source, intensity approximately follows the inverse-square law: doubling distance reduces intensity to one-quarter.

Q206. If distance from a point radiation source doubles, intensity is approximately:

A. Doubled

B. Halved

C. One-quarter

D. Unchanged

Recall note: ALARA means As Low As Reasonably Achievable. Radiation dose is reduced by minimizing time, maximizing distance and using shielding. For a point source, intensity approximately follows the inverse-square law: doubling distance reduces intensity to one-quarter.

Quick Verification References

- OSHA 29 CFR 1926.1053 - Ladders: portable ladder strength and 4:1 setup.
- OSHA 29 CFR 1910.134 - Respiratory Protection: Assigned Protection Factors.
- U.S. EPA - Universal Waste: batteries, pesticides, mercury-containing equipment, lamps and aerosol cans.
- FEMA/USFA ICS materials - Safety Officer monitors safety conditions and develops safety measures.
- OSHA 29 CFR 1910.36-.37 - Exit routes, lighting, alarms and egress requirements.
- OSHA 29 CFR 1910.1001 - Asbestos: 0.1 f/cc 8-hour TWA; 1.0 f/cc 30-minute excursion limit.
- CDC/NIOSH - Revised NIOSH Lifting Equation: RWL and Lifting Index.
- FMCSA 49 CFR Part 382 guidance - random, reasonable-suspicion, post-accident, return-to-duty and follow-up testing.
- PHMSA - UN 2814 Infectious substance, affecting humans; UN 2900 affecting animals only.
- OSHA Powered Industrial Trucks eTool - carry forklift load at lowest practical position, approximately 4-6 inches from ground.